

8085-BASED CHASER LIGHT



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Light chaser circuits are used to create lighting animation sequences for advertising and event promotion (such as the marquees at local movie theatres). They can also be used for decoration and producing pleasing effects for entertainment.

Everyone can afford to build a light chaser, which may be just what you need for your next gathering or party. Most of them have some pre-programmed sequences that change automatically according to your choice, while others are programmed to generate particular patterns like pictures and graphical design.

Here is one such application based on 8085 microprocessor that generates five different lighting effects using bulbs connected to 24 lines of the programmable peripheral interface

PARTS LIST

Semiconductors:

- IC1-IC4 - 74LS04 hex inverter
- IC5 - 7805, 5V regulator
- T1-T24 - BC548 npn transistor
- D1-D4 - 1N4007 rectifier diode
- TRIAC1-TRIAC24 - BT136 triac
- LED1-LED24 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1-R24 - 120-ohm

Capacitors:

- C1 - 1000 μ F, 25V electrolytic
- C2 - 0.1 μ F ceramic disk

Miscellaneous:

- X1 - 230V AC primary to 9V, 500mA secondary transformer
- B1-B24 - 25W, 220V bulb

(PPI) of the microprocessor kit. An 8085 microprocessor kit has at least one 8255 programmable peripheral interface whose ports are extended using suitable connectors for external usage.

Circuit description

Fig. 1 shows the circuit for chaser light based on 8085 microprocessor kit. It comprises inverter 74LS04, transistor BC548, triac BT136, regulator 7805 and a few discrete components. Ports A, B and C of 8255 PPI are configured for mode-'0' operation as output by writing control word 80H to its control register.

The output of the kit's 8255 PPI is inverted by IC 74LS04. The bit information from 74LS04 drives transistor BC548

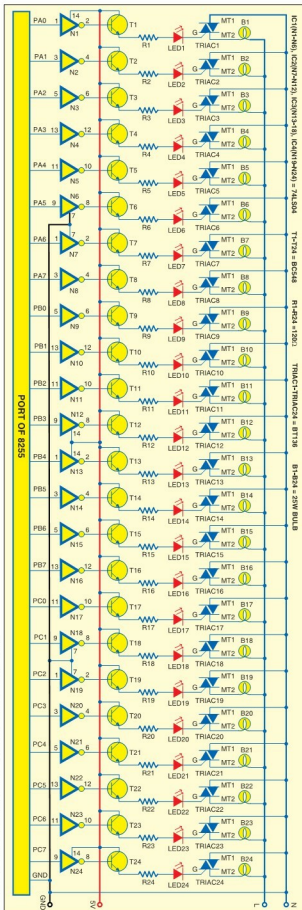


Fig. 1: circuit for chaser light based on 8085 microprocessor kit